

Class Meeting Handout #15

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Hydrolysis of esters in acid or base is the classic example of an acyl transfer reaction. But there is much more. For example, acyl transfer is how amide bonds are made in ribosomes.¹ In our class meeting, we will discuss the hydrolysis of aspirin. This has been extensively studied and we will use it as an inspiration for discussing acid and base catalysis, pH-rate profiles and chemical mechanism.

This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle* Plots were typeset using *Matplotlib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

¹ Read Chapter 10.17 of the textbook

Acyl Transfer Reactions

Discussion: Acyl Groups – 5 minutes

Acyl groups are seen everywhere in biological chemistry and so are a popular subject of physical organic chemistry investigations.

1. Identify the acyl groups below and describe where you would find them in biochemistry? If one is not found in living systems, propose a analogous acyl group that is.

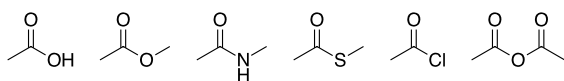


Figure 1: Common acyl groups.

←

2. Comment on the reactivity of the acyl groups above for transfer to water (hydrolysis).

Exercise: Mechanisms of Ester Hydrolysis – 10 minutes

3. Propose the standard mechanism for ester hydrolysis of ethyl acetate in acidic and basic water.
4. Hydrolysis of t-Butyl acetate does not proceed via either of the mechanisms proposed above. It is unable to be hydrolyzed in basic water but is very rapidly hydrolyzed in acidic water. Propose a mechanism. Propose an experiment that would differentiate this mechanism from the usual mechanism for acid-catalyzed hydrolysis.

An Exploration of a Famous pH-rate Profile – 35 minutes

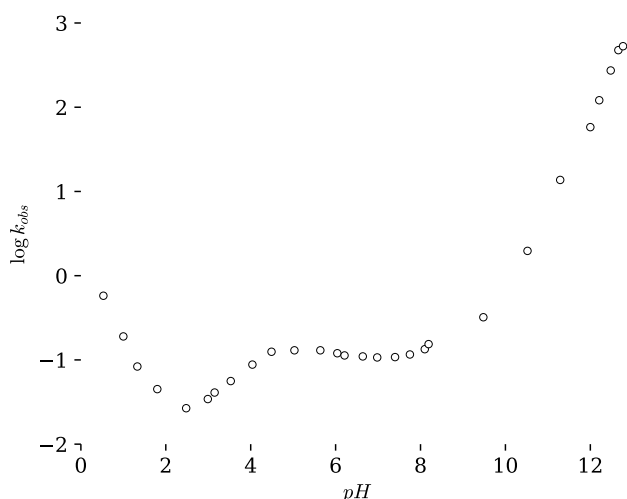
The following exploration discusses a paper in detail. The data is re-analyzed using modern methods. What conclusions will we reach?

Aspirin has been around for a long time.² Its mechanism of action was determined in the 1970's and John Robert Vane shared in the 1982 Nobel Prize in Medicine in recognition of his contribution to that discovery.³

Acetylsalicylic acid (ASA) has a phenolic ester group adjacent to a carboxylic acid group on the benzene ring. The ester can be hydrolyzed in the traditional acid and base-catalyzed pathways. In addition, there is a pathway that involves the carboxylic acid group. Is it acting as an intramolecular base or as a nucleophile? We need to know. This reaction is important in the degradation of aspirin in the bottle, especially children's aspirin that stored as a solution.

A pH-rate Profile

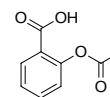
Ester hydrolysis can occur via acid and base-catalyzed pathways. We are familiar with that classic v-shaped pH-rate profile. Let us consider the pH-rate profile for the hydrolysis of the ester group in aspirin presented in figure 3. The data is taken from work by L.J. Edwards⁴ and is presented in table 1.



The rate vs. pH is more complicated than in the case of a simple ester hydrolysis. We clearly see the acid and base-catalysis at the lowest and highest pH values, but what is going on in the middle?

Read ch. 9 and for an example of a pH-rate profile see ch. 10.15

Figure 2: Acetylsalicylic acid. ↓



² "Acetylsalicylic Acid, 2nd ed.", Karsten Schörr, 2016, Wiley-VCH, <https://doi.org/10.1002/9783527685059>, available as an e-book via the UPEI library

³ The Nobel Prize in Physiology or Medicine 1982., *Nobel Prize Outreach AB*, 2022. Accessed on 16 Nov., 2022. <https://www.nobelprize.org/prizes/medicine/1982/summary/>

⁴ "The hydrolysis of aspirin." L.J. Edwards, *Trans. Faraday Soc.*, 1950, 46, 723-735. <https://doi.org/10.1039/TF9504600723>

Figure 3: The pH-rate profile for aspirin hydrolysis.

Table 1: Observed³ first-order rate constants for aspirin hydrolysis. ↓

pH	k_{obs}/days^{-1}	pH	k_{obs}/days^{-1}
0.53	0.578	6.64	0.110
1.00	0.190	6.98	0.107
1.33	0.0835	7.40	0.108
1.80	0.0450	7.75	0.116
2.48	0.0267	8.10	0.134
2.99	0.0343	8.19	0.154
3.15	0.041	9.48	0.321
3.53	0.0561	10.52	1.97
4.04	0.0880	11.29	13.7
4.49	0.125	12.00	58.0
5.03	0.130	12.21	121
5.64	0.130	12.48	273
6.04	0.120	12.66	476
6.21	0.113	12.77	530

Buffer Catalysis

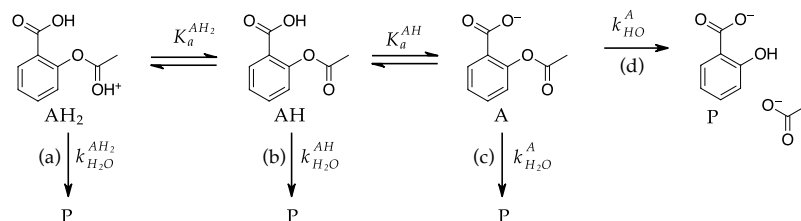
Acetate buffer was used between pH 3 & 6 and the shape of the central "bump" is what is expected if acetate ion is acting as a general base. As acetic acid is titrated to its basic form across the range of its pK_a value, the rate will increase until we reach 100% acetate. However, the buffer was switched to phosphate buffer for the measurements between pH 6 & 8 and we see no significant change in the rate in the pH -independent region that spans pH 5 to 8.

We should confirm if buffer contributes to the rate of catalysis. Fortunately, Edwards had the same thought and performed an experiment where the pH was held constant and the amount of buffer was changed.³ The data and plot are shown in table 2 and figure 4

We see that the rate does not increase with buffer concentration. So in the central region of the pH -rate profile, the shape of the curve is not due to titration the buffer to its basic form. This feature must result from the structure of aspirin. I see a carboxylic acid group in the molecule and aspirin is known to have a pK_a value of 3.6, which is right where the curve of the titration feature occurs in the plot.

The Rate Laws

A proposed reaction scheme for the hydrolysis of aspirin is presented in figure 5 below.



Aspirin has three states of protonation. Each can be the major reacting species depending on the pH . As a result, I can propose four likely routes to product and identify the associated rate constants and K_a values.

- (a) In strongly acid conditions, the carbonyl of the ester can be protonated (AH_2^+) and can be readily attacked by water.

$$\text{The rate law}^5 \text{ is } v_a = k_{H_2O}^{AH_2^+} [AH_2^+]$$

- (b) In weakly acidic conditions, we have the neutral form⁶ of aspirin (AH) that could undergo hydrolysis by water (likely a slow reaction).

$$\text{The rate law is } v_b = k_{H_2O}^{AH} [AH]$$

- (c) Above pH 3.6, aspirin will be mostly in the carboxylate form (A^-). This could also undergo hydrolysis via water as the nucleophile (there is almost

Table 2: Observed³ first-order rate constants vs. concentration of acetate buffer at $pH = 4.6$. ↓

Conc./moles · L ⁻¹	k_{obs}/days^{-1}
0.00521	0.119
0.0104	0.115
0.0209	0.119
0.0417	0.114
0.0833	0.114
0.167	0.116
0.333	0.112

Figure 4: Buffer rate plot. ↓

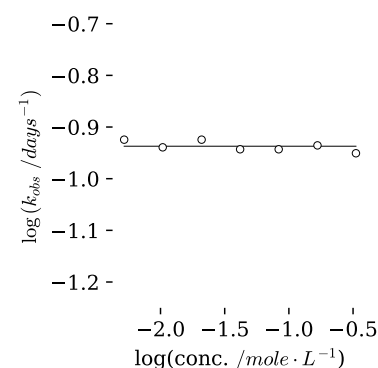


Figure 5: Reaction scheme for aspirin hydrolysis in water.

⁵ Remember that the concentration of the solvent nucleophile, water, is included in the value of $k_{H_2O}^{AH_2^+}$

⁶ Aspirin is a water soluble salt at neutral pH and is a neutral carboxylic acid a lower pH (stomach acid). In this neutral form it can pass easily through a cell membrane and the rate of uptake is greatest in the stomach.

no hydroxide ion to act as a nucleophile at pH 3.6) and it is this pathway that results in the titration rate in the central region of the pH -rate profile.

The rate law is $v_c = k_{H_2O}^{A^-} [A^-]$

- (d) At high pH values direct attack by hydroxide ion predominates via the classic basic hydrolysis mechanism.

The rate law is $v_d = k_{HO^-}^{A^-} [HO^-] [A^-]$

The Interesting Reaction

The rates due to pathways (a), (b) and (d) are the classic ester hydrolysis mechanisms: the acid-catalyzed, water and base-catalyzed rates. The feature in the middle pf the pH -rate profile represents the titration of the benzoic acid group of aspirin. It requires the benzoate (A^-) species but is not using hydroxide as the nucleophile.

The titration region is path (c). Is the basic carboxylate group helping water attack the ester group or is it doing more than that? Figure 6 presents a mechanism where the benzoate group acts as a general base. This is a form of intramolecular catalysis. Figure 7 details a mechanism where the acetate acts as the nucleophile to produce a reactive anhydride intermediate that is rapidly hydrolyzed to the products. This is an example of anchimeric assistance.

Both reactions have the same rate law. We cannot distinguish between them using kinetics. Propose an experiment that would provide evidence for one mechanism over the other. This step is still argued over to this day. There is evidence reported that favour both possibilities.⁷ So even a reaction as simple as the hydrolysis of aspirin can be a puzzle that lasts a century.

Making a Model

We have four rate laws that total up to the observed rate at any given time. Let us combine them into a single rate law that will model the reaction across the entire pH range.

$$v_{obs} = v_a + v_b + v_c + v_d$$

$$v_{obs} = k_{H_2O}^{AH_2^+} [AH_2^+] + k_{H_2O}^{AH} [AH] + k_{H_2O}^{A^-} [A^-] + k_{HO^-}^{A^-} [HO^-] [A^-] \quad (1)$$

Figure 6: Proposed mechanism #1: intramolecular general base catalysis. ↓

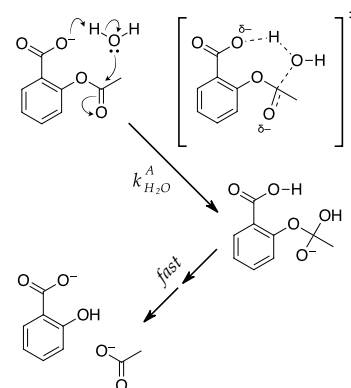
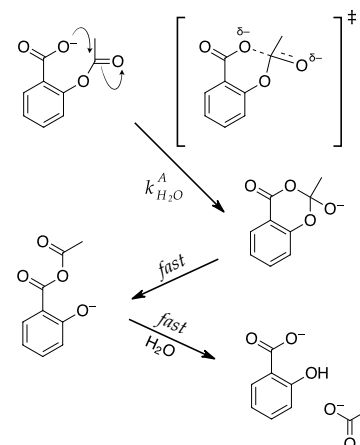


Figure 7: Proposed mechanism #2: anchimeric assistance by carboxylate. ↓



⁷ The evidence clearly favours the anhydride pathway, in my opinion. But the other side just won't give in. Perhaps you will be the one to finally beat them down with definitive experimental evidence. Get to work.

...and we know that...

$$K_a^{AH_2^+} = \frac{[AH][H^+]}{AH_2^+}$$

$$K_a^{AH} = \frac{[A^-][H^+]}{AH}$$

$$K_W = [H^+][HO^-]$$

So we can state the model in terms of $[AH]$ and $[H^+]$...

$$v_{obs} = k_{H_2O}^{AH_2^+} \frac{[H^+]}{K_a^{AH_2^+}} [AH] + k_{H_2O}^{AH} [AH] + k_{H_2O}^{A^-} \frac{K_a^{AH}}{[H^+]} [AH] + k_{HO^-}^{A^-} \frac{K_W}{[H^+]} \frac{K_a^{AH}}{[H^+]} [AH]$$

$$v_{obs} = \left(k_{H_2O}^{AH_2^+} \frac{[H^+]}{K_a^{AH_2^+}} + k_{H_2O}^{AH} + k_{H_2O}^{A^-} \frac{K_a^{AH}}{[H^+]} + k_{HO^-}^{A^-} \frac{K_W}{[H^+]} \frac{K_a^{AH}}{[H^+]} \right) [AH]$$
(2)

We can get $[AH]$ in terms of $[A]_t$ and $[H^+]$.

$$[A]_t = [AH_2^+] + [AH] + [A^-]$$

$$[A]_t = \frac{[H^+]}{K_a^{AH_2^+}} [AH] + [AH] + \frac{K_a^{AH}}{[H^+]} [AH]$$

$$[A]_t = \left(\frac{[H^+]}{K_a^{AH_2^+}} + 1 + \frac{K_a^{AH}}{[H^+]} \right) [AH]$$

...and so...

$$[AH] = \frac{[A]_t}{\frac{[H^+]}{K_a^{AH_2^+}} + 1 + \frac{K_a^{AH}}{[H^+]}}$$
(3)

Plug the eq. 3 into eq. 2 to obtain an equation that describes our model for the system.

$$v_{obs} = \frac{k_{H_2O}^{AH_2^+} \frac{[H^+]}{K_a^{AH_2^+}} + k_{H_2O}^{AH} + k_{H_2O}^{A^-} \frac{K_a^{AH}}{[H^+]} + k_{HO^-}^{A^-} \frac{K_W}{[H^+]} \frac{K_a^{AH}}{[H^+]}}{\frac{[H^+]}{K_a^{AH_2^+}} + 1 + \frac{K_a^{AH}}{[H^+]}} [A]_t$$

The reaction is a first order reaction in $[A]_t$ and can be expressed as...

$$v = k_{obs} [A]_t$$

...where...

$$k_{obs} = \frac{k_{H_2O}^{AH_2^+} \frac{[H^+]}{K_a^{AH_2^+}} + k_{H_2O}^{AH} + k_{H_2O}^{A^-} \frac{K_a^{AH}}{[H^+]} + k_{HO^-}^{A^-} \frac{K_W}{[H^+]} \frac{K_a^{AH}}{[H^+]}}{\frac{[H^+]}{K_a^{AH_2^+}} + 1 + \frac{K_a^{AH}}{[H^+]}}$$

We can say that $K_a^{AH_2^+} \gg [H^+]$ because the protonated ester is a very strong acid. So the equation that will fit the data in the pH range of the experiment can be simplified to...

$$k_{obs} = \frac{k_{H_2O}^{AH_2^+} \frac{[H^+]}{K_a^{AH_2^+}} + k_{H_2O}^{AH} + k_{H_2O}^{A^-} \frac{K_a^{AH}}{[H^+]} + k_{HO^-}^{A^-} \frac{K_W}{[H^+]} \frac{K_a^{AH}}{[H^+]}}{1 + \frac{K_a^{AH}}{[H^+]}} \quad (4)$$

Fitting the Data

The equation required fitting a data set to seven parameters. For at least one of the parameters there was not enough information to enable a convergence to an answer. We can approach this piecemeal instead. Each region of the graph is dominated by a given pathway in a given pH domain.

In the acid catalyzed region below pH 2 we can say that only the $k_{H_2O}^{AH_2^+}$ term is important and we could simplify the equation for k_{obs} to...

$$k_{obs}^a = \frac{k_{H_2O}^{AH_2^+} \frac{[H^+]}{K_a^{AH_2^+}}}{1 + \frac{K_a^{AH}}{[H^+]}}$$

...and we are also at a pH value well below the value of K_a^{AH} , so we can say that $[H^+] \gg K_a^{AH}$ and simplify the equation for k_{obs} when $pH < 2$ to...

$$k_{obs}^{pH < 2} = \frac{k_{H_2O}^{AH_2^+}}{K_a^{AH_2^+}} [H^+] \quad (5)$$

The slope of the four data points below pH 2 is $k_{H_2O}^{AH_2^+}/K_a^{AH_2^+}$. We can treat all four terms that way. There is enough data in the plot to separately fit terms for $k_{H_2O}^{AH_2^+}/K_a^{AH_2^+}$ ($pH < 2$), $k_{H_2O}^{A^-}$ ($3 < pH < 8$), and $k_{HO^-}^{A^-}$ ($pH > 10$). There is no region that is clearly dominated by the $k_{H_2O}^{AH}$ term and it may be so slow that it does not matter in the overall observed value for k_{obs} .⁸ We will ignore that term for now. Below are the equations used to fit the other two regions of the plot. Derive them yourself and check my math.

$$k_{obs}^{pH 3-8} = \frac{k_{H_2O}^{A^-} \frac{K_a^{AH}}{[H^+]}}{1 + \frac{K_a^{AH}}{[H^+]}} = \frac{k_{H_2O}^{A^-} K_a^{AH}}{[H^+] + K_a^{AH}} \quad (6)$$

$$k_{obs}^{pH > 10} = \frac{k_{HO^-}^{A^-} \frac{K_W}{[H^+]} \frac{K_a^{AH}}{[H^+]}}{1 + \frac{K_a^{AH}}{[H^+]}} = k_{HO^-}^{A^-} \frac{K_W}{[H^+]} \quad (7)$$

⁸ See the Jupyter notebook that accompanies this document for documentation on the methods for fitting the regions of the plot to each of the three expressions.

Figure 8 presents the data with three separate curve fits according to eq. 5, 6 & 7. We know that $K_W = 10^{-14}$. The extra curve on the graph is the term due to $k_{H_2O}^{AH}$. I placed it arbitrarily. Do you think it is needed to accurately fit the data to the model?

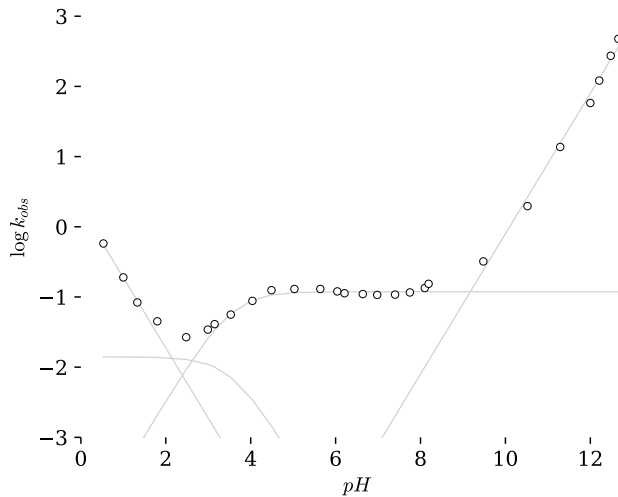


Figure 8: The pH-rate profile for aspirin hydrolysis with proposed reaction curves.
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Values from the piecemeal curve fits.

$$\begin{aligned} k_{H_2O}^{AH_2^+}/K_a^{AH_2^+} &= 1.88 \text{ M}^{-1} \text{ days}^{-1} \\ k_{H_2O}^{AH} &= \text{n/a} \\ k_{H_2O}^{A^-} &= 0.119 \text{ days}^{-1} \\ k_{HO^-}^{A^-} &= 7970 \text{ M}^{-1} \text{ days}^{-1} \\ K_a^{AH} &= 10^{-3.55} \end{aligned}$$

Using the piecemeal curve fits, we extracted the values for the rate constants. I also allowed the curve fit to optimize for the value of K_a^{AH} . The result was very close to the reported literature value and I used my calculated value in the next steps.

Now we have most of the parameters. Let us plot the system assuming that the $k_{H_2O}^{AH}$ term is unimportant. I modified the model for the system by including all the values that have been found into eq. 4 on the previous page and setting $k_{H_2O}^{AH} = 0$. This model is stated in eq. 8. The calculated line is plotted with the experimental data in figure 9 on the next page. We see that there is something missing. The curve dips below the data in the region where we anticipate a possible contribution from the $k_{H_2O}^{AH}$ pathway. I then curve fit the model to optimize for the value of $k_{H_2O}^{AH}$. The optimal curve fit was found when $k_{H_2O}^{AH} = 0.0107$.

$$k_{obs} = \frac{1.88[H^+] + k_{H_2O}^{AH} + 0.119 \frac{10^{-3.55}}{[H^+]} + 7970 \frac{10^{-14}}{[H^+]} \frac{10^{-3.55}}{[H^+]}}{1 + \frac{10^{-3.55}}{[H^+]}} \quad (8)$$

I then fit the complete model described in eq. 4 on the previous page and allowed all the parameters to optimize. This curve fit with the optimized values for the parameters $k_{H_2O}^{AH_2^+}/K_a^{AH_2^+}$, $k_{H_2O}^{AH}$, $k_{H_2O}^{A^-}$, $k_{HO^-}^{A^-}$, and K_a^{AH} are also shown in figure 9 on the next page.

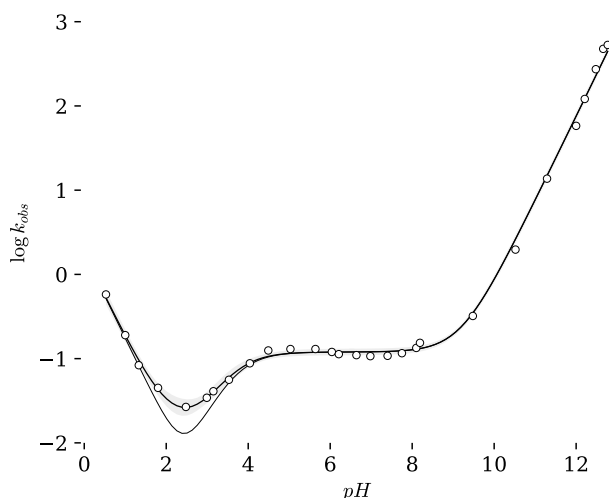
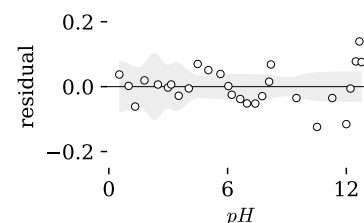


Figure 9: The pH-rate profile for aspirin hydrolysis with curve fits. The fit without the $k_{H_2O}^{AH}$ term is the thin line; the curve fit optimizing for $k_{H_2O}^{AH}$ is the thicker line.

Figure 10: Residual plot. ↓



Values from the constrained curve fit.

$$\begin{aligned} k_{H_2O}^{AH+}/K_a^{AH_2^+} &= (1.75 \pm 0.17) \text{ M}^{-1} \text{ days}^{-1} \\ k_{H_2O}^{AH} &= (0.0142 \pm 0.0045) \text{ days}^{-1} \\ k_{H_2O}^{A-} &= (0.120 \pm 0.005) \text{ days}^{-1} \\ k_{HO}^{A-} &= (7560 \pm 410) \text{ M}^{-1} \text{ days}^{-1} \\ K_a^{AH} &= 10^{-3.65 \pm 0.12} \end{aligned}$$

⁹ Accuracy depends on the model. I used a value for the water ion product of $K_W = 10^{-14}$ in the curve fit that gave the above parameters. However, the authors performed their experiment at 17 °C. See the *Python* notebook for notes on calculating the value of K_W and the new curve fit using a value of $K_W = 10^{-14.26}$. Did it make a difference?

A Final Note on Curve Fitting

I was finally able to fit the model to the data with all parameters being optimized. I had to set constraints to keep the parameters from drifting into the vast numeric space available when one is fitting a model with five variables. The result of fitting all five variables gave similar values to the piecemeal method but will be more accurate⁹ since each value is now dependant on many more data points. Using a computational tool that handles uncertainties I was able to easily extract the standard deviations for each parameter from the curve fit result. Observe the high uncertainty for the $k_{H_2O}^{AH}$ value (almost a 32% error).

Conclusion

We have examined the mechanism for the hydrolysis of aspirin and identified four different pathways that contribute to the rate of hydrolysis at various values of pH. A rate law was constructed to model the reaction and the parameters for three of the pathways were determine directly from regions of the plot where each dominated. The final term was optimized using the model for the curve fit and parameters from the three pathways that were fitted from discrete regions of the plot. All four pathways were observed to have significant contributions to the curve fit of the model.

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will...

- have reviewed the idea of catalysis in reactions and understand the concept of effective molarity.
- have reviewed the definitions of specific and general acid/base catalysis.
- be able to derive a rate law for specific and general acid/base catalysed reactions.
- be able to interpret pH-rate profile.
- be able to derive a mathematical model that explains an observed pH-rate profile.
- be able to use the pH-rate profile data to obtain rate constants and equilibrium constants for a system.
- **Computer Skill:** be able to use an interactive *Python* notebook to perform and document curve fits of experimental data to mathematical models.

Appendix

Here is a reminder of the definitions for general and specific acid or base catalysis.

ACID CATALYSIS is when a reaction is accelerated by a proton that is donated to the molecule and returned by end of the process. Base catalysis is when a proton is accepted from the reactant during the reaction and given back by the end. The total concentration of acid or base does not change as the reactant is converted to product.

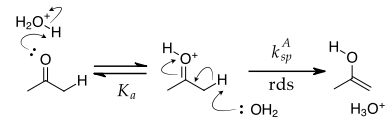
SPECIFIC ACID CATALYSIS is when the acidic conditions are strong enough so that a protonated form of the reactant exists in an equilibrium. This intermediate then proceeds to product via a rate-determining step (rds). Specific base catalysis has deprotonation as an equilibrium step before a rds that produces product. In both cases, the amount of intermediate depends on the protonating power of the system, often expressed as pH or H_0 .

GENERAL ACID CATALYSIS is when an acid is not strong enough to protonate the reactant before the rds. The reactant becomes basic enough to accept the proton as the reaction proceeds. The proton transfer takes place during the rds and the acid is part of the transition state. General base catalysis is when the base is part of the rds. This is a second-order process and the rate will depend on the concentration of the acid/base involved. There are many acids and bases in the world other than hydronium or hydroxide ions.

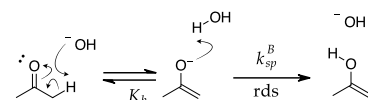
Read ch. 9 and for an example of a pH-rate profile see ch. 10.15

Figure 11: Definitions for acid/base catalysis using the tautomerization of a ketone/enol pair as an example. (See question 9.10) ↓

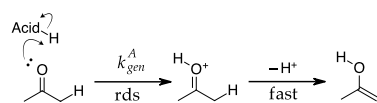
Specific acid catalysis



Specific base catalysis



General acid catalysis



General base catalysis

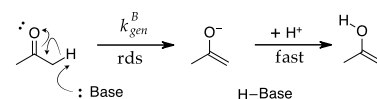


Table 3: Rate laws for the four cases described in figure 11. ↓

Specific acid	Specific base	General acid	General base
$A + H^+ \xrightleftharpoons{K_a} AH^+ \xrightarrow{k_{sp}^A} P$	$A + HO^- \xrightleftharpoons{K_b} A^- \xrightarrow{k_{sp}^B} P$	$A + BH \xrightarrow{k_{gen}^A} P$	$A + B \xrightarrow{k_{gen}^B} P$
$v = \frac{[H^+]}{K_a + [H^+]} k_{sp}^A [A]_t$	$v = \frac{[HO^-]}{K_b + [HO^-]} k_{sp}^B [A]_t$ or $v = \frac{K_b}{K_a + [H^+]} k_{sp}^B [A]_t$	$v = k_{gen}^A [BH] [A]_t$ or $v = \frac{[H^+]}{K_a^{BH} + [H^+]} k_{gen}^A [B]_t [A]_t$	$v = \frac{[H^+]}{K_a + [H^+]} k_{gen}^A [A]_t$ or $v = \frac{K_a^{BH}}{K_a^{BH} + [H^+]} k_{gen}^B [B]_t [A]_t$
$k_{obs} = \frac{[H^+]}{K_a + [H^+]} k_{sp}^A$	$k_{obs} = \frac{K_b}{K_a + [H^+]} k_{sp}^B$	$k_{obs} = \frac{[H^+]}{K_a^{BH} + [H^+]} k_{gen}^A [B]_t$	$k_{obs} = \frac{K_a^{BH}}{K_a^{BH} + [H^+]} k_{gen}^B [B]_t$

PRACTICE YOUR MATH SKILLS. Rate laws can be stated for the reactions defined in figure 11 and are listed in table 3. Show how each rate law was derived to give a term for k_{obs} that is dependent on the acidity of the solution.